温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3101 W03 Foundations of Business Analytics

Statistical Inference I: Sampling and Estimation

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Outline

- Probability Distribution
  - Basic concepts of probability
  - Random variables
  - Discrete and continuous probability distribution
- Sampling and Estimation
  - Sampling
  - Point estimation
  - Sampling distribution

Basic Concepts of Probability

- Probability is the likelihood that an outcome occurs.
  - An experiment is the process that results in an outcome.
  - The outcome of an experiment is a result that we observe.
  - The sample space is the collection of all possible outcomes of an experiment.
  - An **event** is **a collection of one or more outcomes from a sample space**.
- Two basic facts govern probability
  - Probability associated with any outcome must be between 0 and 1.

$$0 \leq P(O_i) \leq 1 \text{ for each outcome } O_i$$

  - Sum of the probabilities over all possible outcomes must be equal to 1.

$$P(O_1) + P(O_2) + \cdots + P(O_n) = 1$$

(n is the number of outcomes in the sample space.)

Random Variables

- A random variable is a numerical description of the outcome of an experiment.
  - A **discrete** random variable is one for which the number of possible outcomes can be counted.
    - E.g. outcomes of dice rolls, whether a customer likes or dislikes a product, number of hits on a Web site link today
  - A **continuous** random variable has outcomes over one or more continuous intervals of real numbers.
    - E.g. weekly change in DJIA (Dow Jones Industrial Average), daily temperature, time between machine failures
- Probability distribution: a characterization of the possible values that a random variable may assume along with the probability of assuming these values.

Discrete Probability Distributions

- Probability **Mass** Function (pmf)
  - A mathematical function $f(x)$ specifying the probability of the random variable X.
    - x_i represents the i th value of X.
    - Properties of a pmf:

$$0 \leq f(x_i) \leq 1 \text{ for all } i$$

$$\sum_i f(x_i) = 1$$

- Cumulative Distribution Function (cdf):

$$F(x) = P(X \leq x)$$

Continuous Probability Distributions

- Probability **Density** Function (pdf)
  - A curve described by a mathematical function that characterizes a continuous random variable
    - Properties of a pdf:
      - $f(x) \geq 0$ for all values of x
      - **<u>Total area</u>** under the density function equals 1.
      - Probability are only defined over an interval.
      - $P(a \leq X \leq b)$ is the **<u>area</u>** under the density function between a and b.
- Cumulative Distribution Function (cdf):

$$F(x) = P(X \leq x)$$

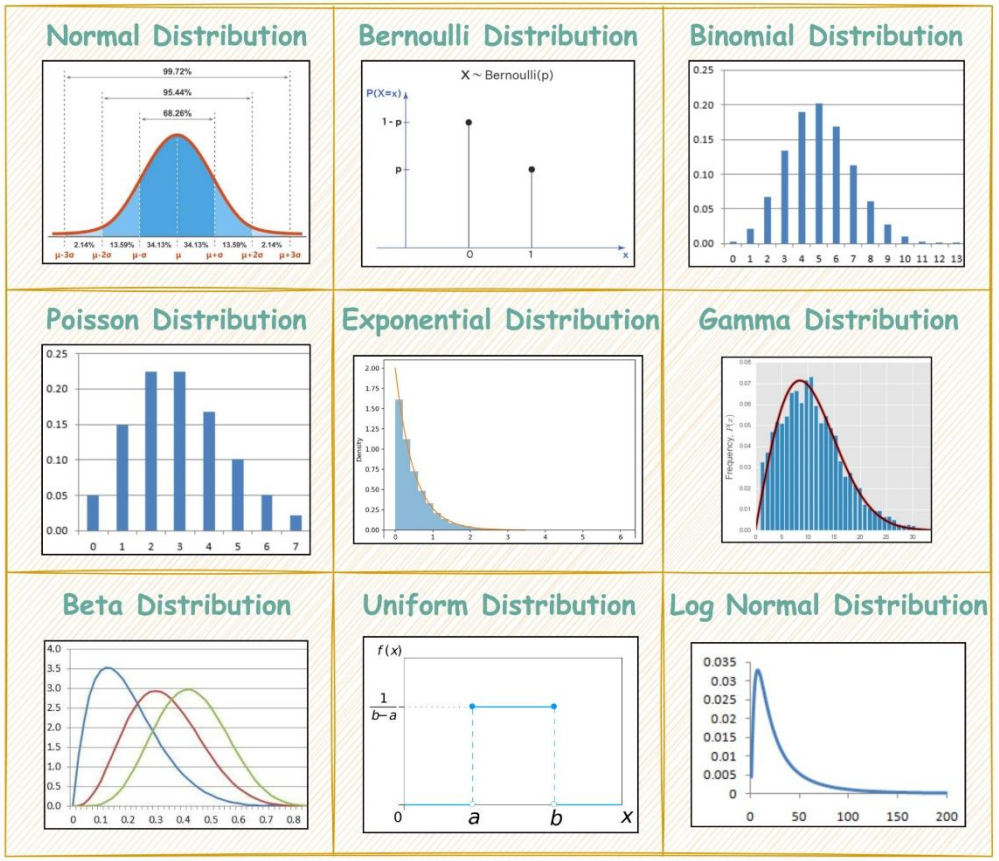
$$P(a \leq X \leq b) = P(X \leq b) - P(X \leq a) = F(b) - F(a)$$

Commonly Used Distributions

- Discrete Variables
  - Bernoulli Distribution
  - Binomial Distribution
  - Poisson Distribution

    …
- Continuous Variables
  - Uniform Distribution
  - Normal Distribution
  - (Student's) t-distribution
  - Exponential Distribution

    …



https://blog.dailydoseofds.com/p/nine-most-important-distributions

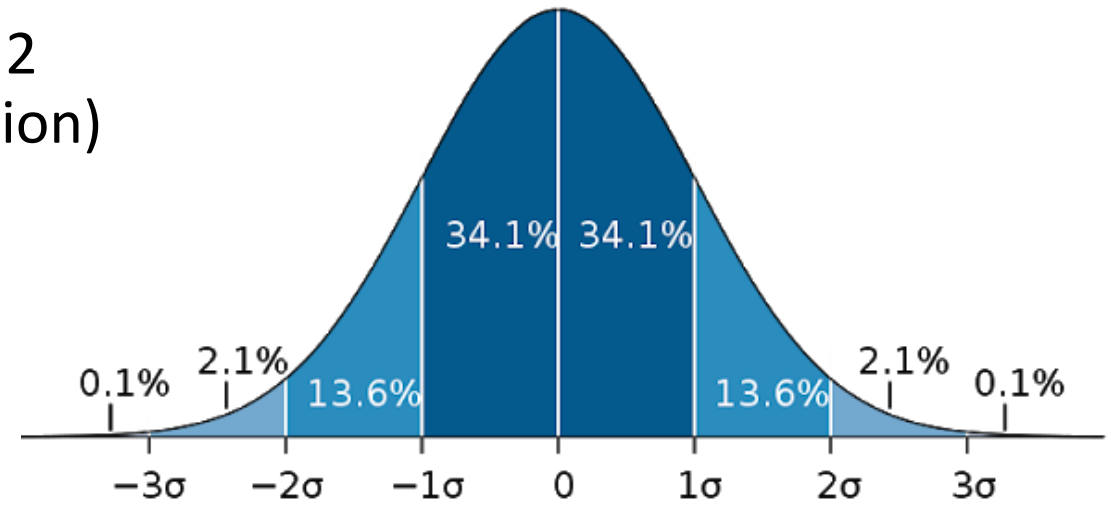
Question: Why do we need to know about these distributions?

Normal Distribution

- $f(x)$ is a bell-shaped curve characterized by 2 parameters μ (mean) and σ (standard deviation)

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Properties
  - Symmetric, Mean = Median = Mode
  - Range of X is unbounded
  - Empirical rules apply (i.e., the area under the density function within ± 2 standard deviation is 95.4%, and that within ±3 standard deviation is 99.7%)



Motivating Example

How Much Do WKU Fresh Graduates Earn?

- Each year, hundreds of WKU students graduate and start their first jobs. The Career Service Office wants to know:

  💰 *What is the **average monthly salary** of WKU graduates in their first job?*

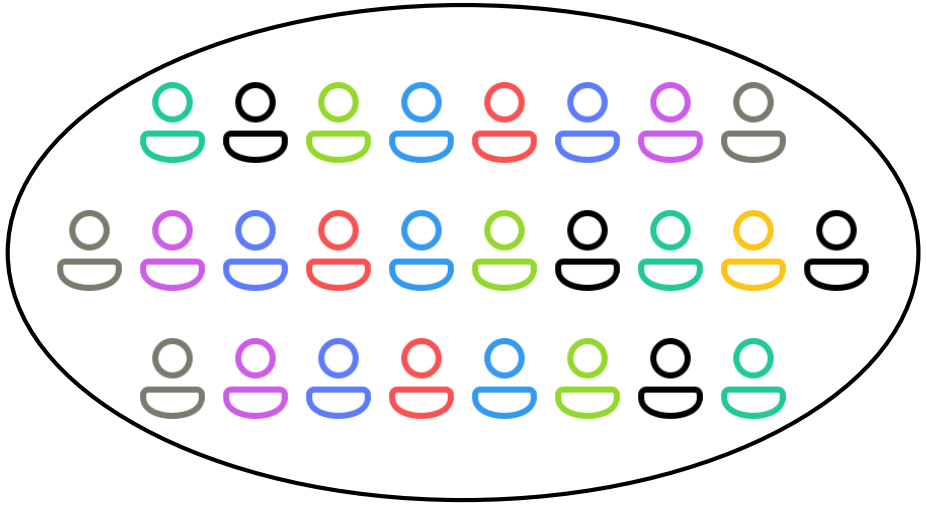
- However, it is unrealistic to survey every graduate. Why?
- To make it practical, suppose we surveyed **120** graduates, and their average monthly salary is $\underline{\mathbf{¥25,000}}$.
- Questions to think about:
  - Does this sample reflect all WKU graduates?
  - How accurate is this estimate?
  - What if our sample happened to include mostly business majors or top performers?

What This Example Tells Us

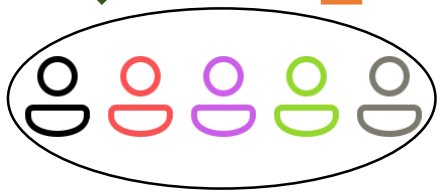
- This example raises two key questions:
  - **Sampling:** How can we select a sample that *truly represents* the whole population?
  - **Estimation:** How can we use that sample to *infer* the *true* population characteristics?
- From **a small sample**, we want to learn about **a large population** – but every sample gives a slightly different result.
- Therefore, we need:
  - **Good sampling methods** to minimize bias
  - **Statistical estimation** to measure uncertainty (e.g., confidence intervals)
- Think about it:
  - If we repeated the survey with *another* 120 graduates, would we get the same average salary? Why or why not?
  - And how do we calculate the *average monthly salary* now?

Sampling and Estimation

A Population (N)



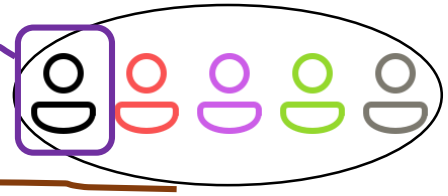
Sampling **Inference**



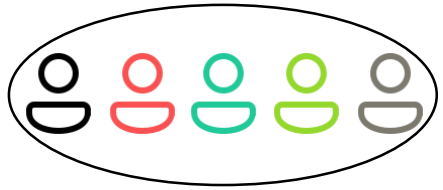
A Sample (n)

What if we have many samples?

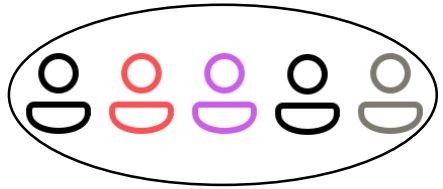
A sample vs An observation



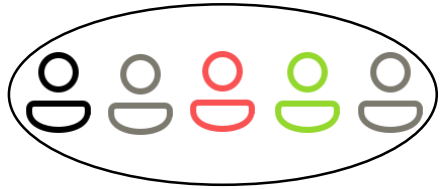
Sample 1: $(\bar{x}_1, s_1)$



Sample 2: $(\bar{x}_2, s_2)$



Sample 3: $(\bar{x}_3, s_3)$



Sample 4: $(\bar{x}_4, s_4)$

Population parameter

- Mean: $\boldsymbol{\mu}$
- Standard deviation: $\boldsymbol{\sigma}$

Questions:

- Is $\bar{x}$ a good estimator of μ?
- Is s a good estimator of σ?
- ***What is a good estimator?***

Sample estimate

- Mean: $\bar{\boldsymbol{x}}$
- Standard deviation: $\boldsymbol{s}$

Sampling – What, Why, How

- What is sampling?
  - Sampling is the process of selecting a **subset (sample)** from a **larger group (population)** to draw conclusions about the whole.
  - It is the foundation of statistical analysis and data-driven decision making.
- Why do we use sampling instead of a census?

  💰 **Cost:** Conducting a full census is often too expensive.

  ⏰ **Time:** A census can be time consuming; results may be needed more quickly.

  🔄 **Change population:** When populations change rapidly, census data can become outdated before completion.

  🎯 **Sufficiency:** If approximate information is acceptable, a census may be unnecessary.

  🌐 **Destructive Testing:** In some cases (e.g., product testing), collecting data from all items would destroy the population.

Sampling – What, Why, How

- How? – Sampling Methods

1) Simple Random Sampling
   - Each possible sample of size n from a finite population of size N has an equal probability of being selected.
2) Random Sampling
   - A random sample of size n is selected from an infinite population so that
     - Each element selected comes from the **same population**
     - Each element is selected **independently** $\rightarrow$ prevent **selection bias**

Two Key Points about Population

- Population is a relative concept — it depends on the research question, focus, or purpose of the study.
- **Ongoing processes** (e.g., customer arrivals, product orders, or web clicks) often create an **infinite** population, since there is **no fixed upper limit** on the number of possible observations.

Example: Random Sampling from an Infinite Population

- Scenario
  - A restaurant wants to understand its customers. An employee wants to select and interview a sample of arriving customers.
- Think About It:
  - What is the population in this case?
  - Why can we treat it as **infinite**?
  - Who should be included — and who should be excluded — in the sample*? (e.g., should someone who just uses the restroom count?)*
  - How can the employee ensure each customer is selected independently?
  - What kinds of choices could introduce selection bias? *(e.g., choosing only certain age groups or groups of friends)*
  - How can we design a fair, random sampling process here?

<u>Key Takeaway:</u> In an ongoing process like customer arrivals, we can approximate an infinite population —but randomness and independence are critical to avoid bias.

Sampling – What, Why, How

- Other Probability Sampling Methods
  - **Stratified sampling:** divide the population into homogeneous subgroups (strata), then take a random sample from each.

    → Improves accuracy when subgroups differ (e.g., gender, major, income level).
  - **Cluster sampling:** divide the population into heterogeneous groups (clusters), then randomly select some clusters for analysis.

    → Useful when population is large and geographically dispersed.
  - **Systematic sampling:** arrange the population in a list, select one of the first k elements at random, then choose every kth element thereafter.

    → Simple and efficient, but avoid periodic patterns.
- Non-Probability Sampling (when randomness is not guaranteed)
  - **Convenience sampling:** select who is easiest to reach.
  - **Judgment sampling:** select based on researcher's opinion or expertise.

Sampling – What Is a Good Sample?

- A good sample is one that is representative – it resembles the population from which it was drawn.
- Two types of sampling errors to consider:

⚙ **Sampling (statistical) error**

- Occurs because a sample is only a subset of the population.
- Depends on sample size relative to the population.
- Cannot be completely avoided — only reduced by larger, random samples.

🚫 **Non-sampling error**

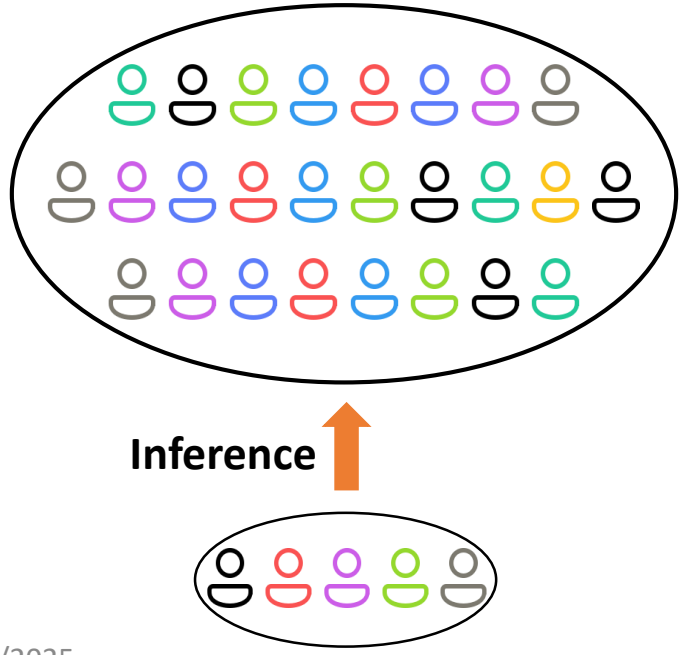
- Occurs when the sample fails to represent the target population.
- Often results from poor design or wrong population frame (e.g., convenience sampling).

Think About It:

- What does "similar to the population" really mean?
- Which sample better represents all WKU graduates?
  - A: 100 business students
  - B: 25 students from each major
- What can make a sample **unrepresentative**?

Statistical Inference – Point Estimation

- A sample statistic serves as a point estimator of a population parameter.
- An estimator is a random variable — its value depends on the sample data.
- An estimate is the numerical value computed from the sample.
- A point estimate gives a single number, unlike an interval estimate, which provides a range of plausible values.



Population parameter
- Mean: $\boldsymbol{\mu}$
- Standard deviation: $\boldsymbol{\sigma}$

Sample estimate
- Mean: $\bar{\boldsymbol{x}}$
- Standard deviation: $\boldsymbol{s}$

$$\mu = \frac{\sum_{i=1}^{N} x_i}{N} \qquad \sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N-1}}$$

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} \qquad s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

$$\bar{x} = 15 \qquad s = 3$$

Question: Are $\bar{x}$ and s good estimators?

Statistical Inference – What Is a Good Estimator?

- A good estimator provides estimates that are **accurate on average** and **consistent** with the population parameter.
  - Sample mean is a good point estimate for population mean $\rightarrow$ it is unbiased.
  - Sample variance uses $(n-1)$ in the denominator (instead of n) to make it an unbiased estimator of the population variance.
- Unbiased estimator
  - An estimator is unbiased if its expected value equals the population parameter.

$$\boldsymbol{E}[\widehat{\boldsymbol{\theta}}] = \boldsymbol{\theta}$$

- Beyond unbiasedness, other desirable qualities of a "good" estimator include *efficiency*, *consistency*, and *sufficiency*.

Statistical Inference

- In inferential statistics, we distinguish between:
  - The **target population** — the group we want to make inferences about.
  - The **sampled population** — the group from which the sample is actually drawn.
- We use a sample statistic (from the sampled population) as a point estimator of a parameter in the target population.
- In practice, achieving a close match between the sampled and target populations is not always possible.
- When the two populations differ significantly, the sample may not represent the target population accurately — leading to biased or misleading inferences.

Statistical Inference – Sampling Distributions of $\bar{x}$

- If we repeatedly draw random samples of size n from a population N, each sample will produce a different sample mean ($\bar{x}$).
- The sampling distribution of $\bar{x}$ is the probability distribution of this random variable $\bar{x}$ (i.e., $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_m$) $\rightarrow$ It shows how $\bar{x}$ varies from sample to sample.
- Remember
  - $\bar{x}$ $\rightarrow$ a random variable (point estimator)
  - $\bar{x} = 12{,}500$ $\rightarrow$ a specific estimate from one sample
- Why does it matter?
  - The sampling distribution allows us to make probability statements about how close the sample mean ($\bar{x}$) is to the true population mean μ.

Think about it:

- What factors affect how much $\bar{x}$ varies from sample to sample?

Statistical Inference – Sampling Distribution of $\bar{x}$

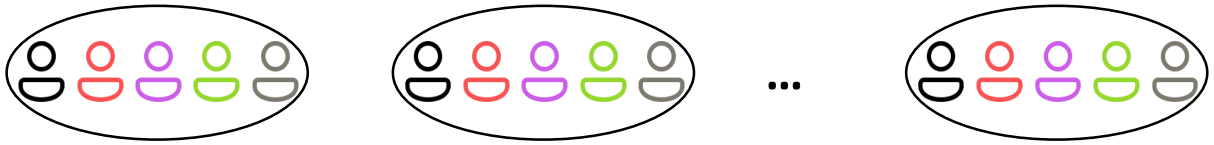
Expected Value of $\bar{x}$

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$\bar{x}$: $(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_m)$

For a simple random sample: $\boldsymbol{E[\bar{x}] = \mu}$

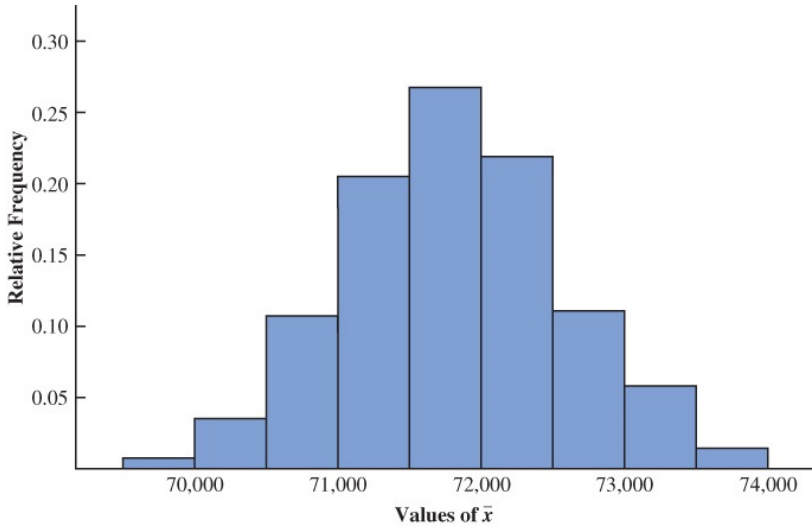
Sample size m



sample 1 $(\bar{x}_1, s_1)$ sample 2 $(\bar{x}_2, s_2)$ … sample m $(\bar{x}_n, s_n)$

Question:

- What is n? what is m?
- Will the choice of n and m influence the unbiasedness?
- Will $\bar{x}$ be unbiased for different sampling methods?



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A bell-shaped appearance centered around their respective population parameter

Statistical Inference – Sampling Distribution of $\bar{x}$

Standard Deviation of $\bar{x}$

- The standard deviation of the of the random variable $\bar{x}$ is the standard error of the mean, $\sigma_{\bar{x}}$. For a *finite population* of size N, and sample size n, we have:

$$\sigma_{\bar{x}} = \sqrt{\frac{N-n}{N-1}}\left(\frac{\sigma}{\sqrt{n}}\right)$$

- Where $\sqrt{(N-n)/(N-1)}$ is the finite population correction factor.
- When $n/N < 0.05$, we can neglect the population correction factor, and $\sigma_{\bar{x}}$ simplifies to the standard error of the mean for an *infinite population*.

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Statistical Inference – Sampling Distribution of $\bar{x}$

Estimated Standard Deviation of $\bar{x}$

- When the population standard deviation **σ is unknown**, we use the sample statistic s as a point estimator of the standard deviation of $\bar{x}$.

Finite Population

$$s_{\bar{x}} = \sqrt{\frac{N-n}{N-1}}\left(\frac{s}{\sqrt{n}}\right)$$

$$\sigma_{\bar{x}} = \sqrt{\frac{N-n}{N-1}}\left(\frac{\sigma}{\sqrt{n}}\right)$$

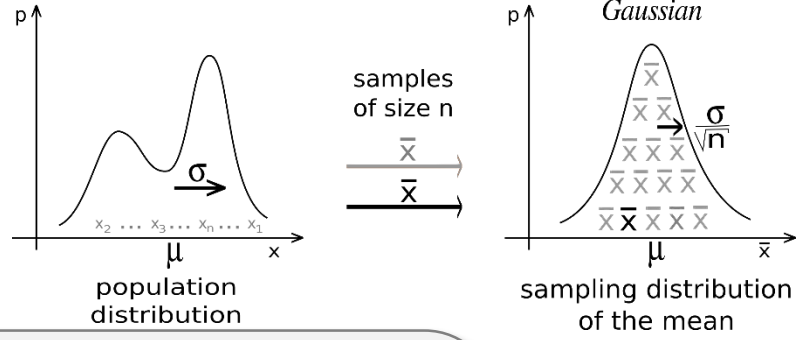
Infinite Population

$$s_{\bar{x}} = \frac{s}{\sqrt{n}}$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

- The difference between $s_{\bar{x}}$ and $\sigma_{\bar{x}}$ is due to **sampling error**.

Central Limit Theorem (CLT)



Central Limit Theorem (Lindeberg–Lévy)

Suppose $X_1, X_2, X_3 \ldots$ is a sequence of i.i.d random variable with $E[X_i] = \mu$ and $Var[X_i] = \sigma^2 < \infty$. Then, as n approaches infinity, the random variables $\sqrt{n}(\bar{X}_n - \mu)$ converge in distribution to a normal $N(0, \sigma^2)$:

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$$

- If the <u>sample size is large enough</u>, then the sampling distribution of the sample mean $\bar{x}$:
  - is <u>*approximately*</u> normally distributed <u>*regardless*</u> <u>of the distribution of the population</u>
  - has a mean equal to the population mean
- If the population is normally distributed, then the sampling distribution is also normally distributed for <u>*any* sample size</u>.
- Why do we care?
  - The CLT allows us to use <u>*the theory we learned about calculating probabilities for normal distributions*</u> to draw conclusions about sample means.

Three Different Population Distributions

Let us consider three different population distributions.

- Population I: uniform distribution
- Population II: rabbit-eared distribution
- Population III: exponential distribution (right-skewed)

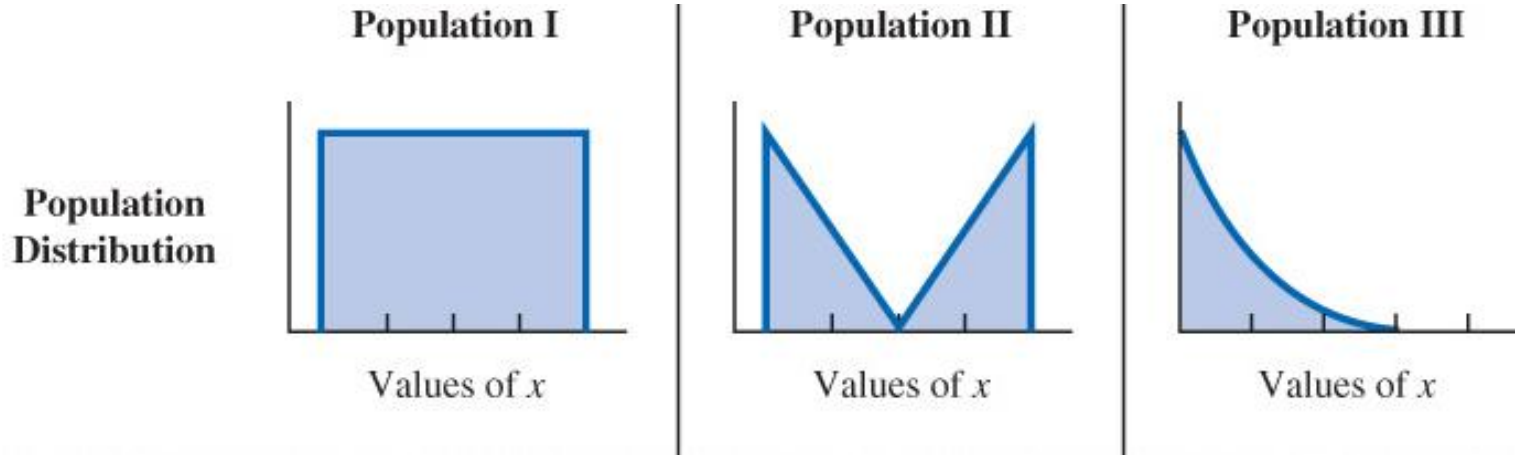


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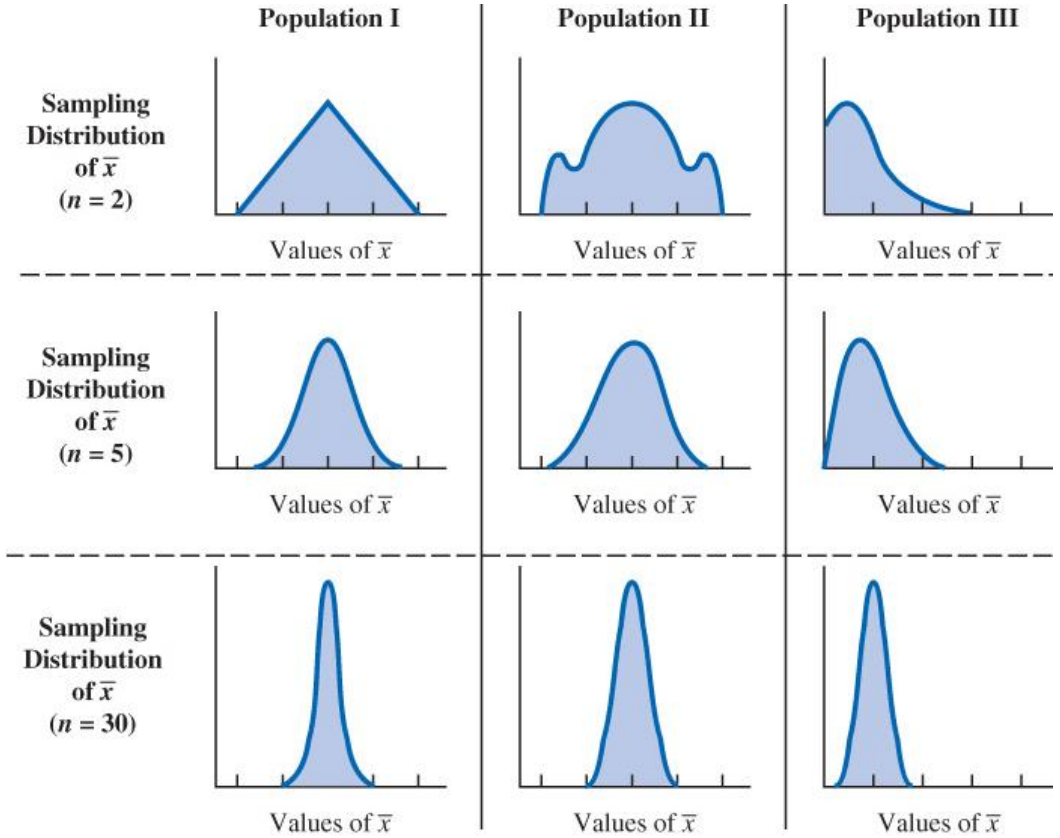
Illustration of the CLT

The three panels shows *what happens to the sampling distribution of $\bar{x}$ for the three population distributions when the sample size n is increased*.

$\mathrm{n} = 2$ sampling distribution shapes differ from the corresponding population distribution shapes.

$\mathrm{n} = 5$ shapes of sampling distributions I and II begin to look similar, but III is still skewed to the right.

$\mathrm{n} = 30$ shapes of each sampling distribution has become approximately normal.



Summary

📘 Today's Key Takeaways

- Sampling: Selecting a representative subset from a population.
- Good Samples: Minimize sampling and non-sampling errors.
- Estimators: Sample statistics used to estimate population parameters.
- Good Estimators: Unbiased, efficient, consistent, and sufficient.
- Sampling Distribution: Describes how a statistic (e.g., $\bar{x}$) varies across samples.
- Central Limit Theorem: Allows us to apply normal probability theory to sample means.

📅 Next Class

- Interval estimation
- Confidence intervals
- Hypothesis testing